

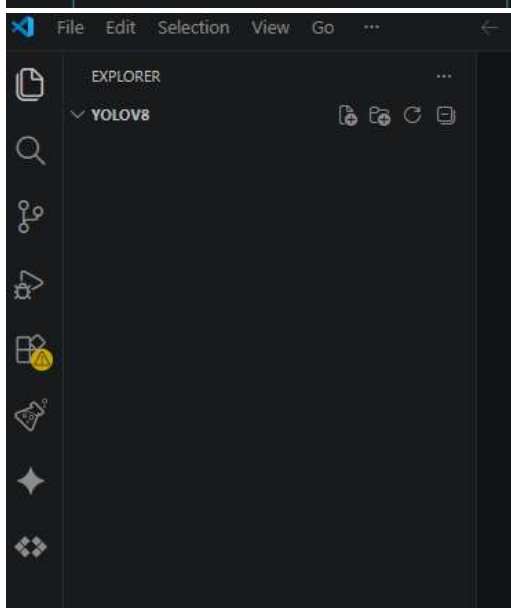
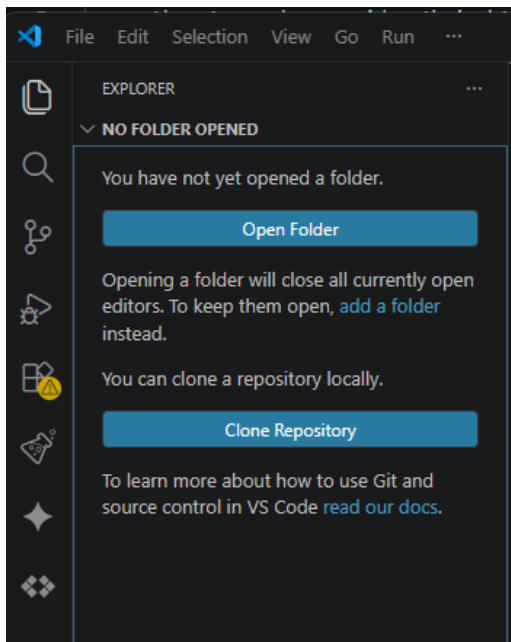
DETEKSI KECEPATAN KENDARAAN DENGAN AI YOLOv8

1. Instal apk virtual studio code
2. Instal extensions python

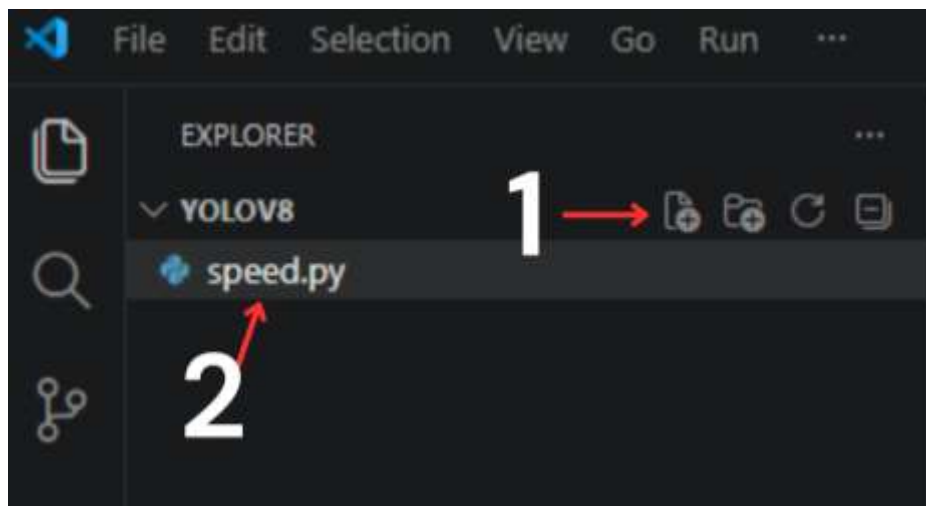


Penulisan program

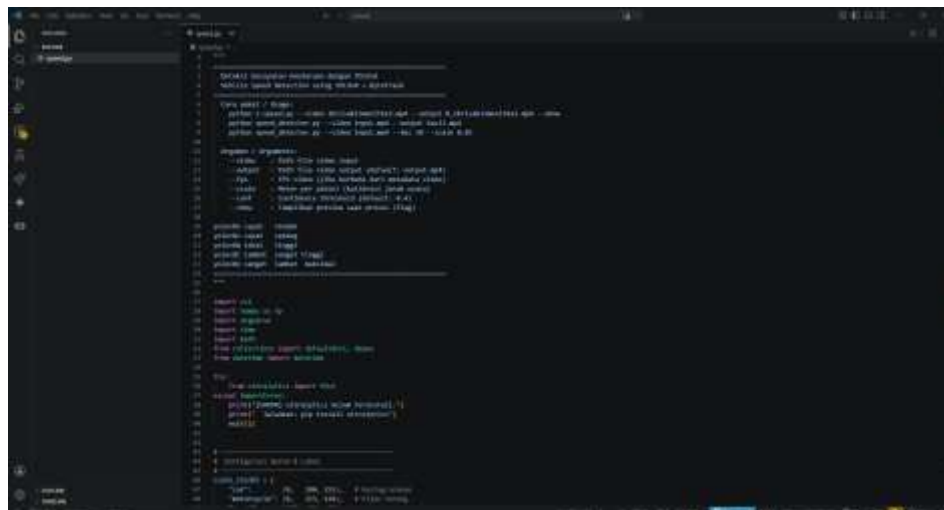
1. Buat new folder dengan nama yolov8 di disk(D)
2. Buka apk virtual studio code – open folder – pilih folder yang dibuat tadi – select folder



3. Buat file speed.py

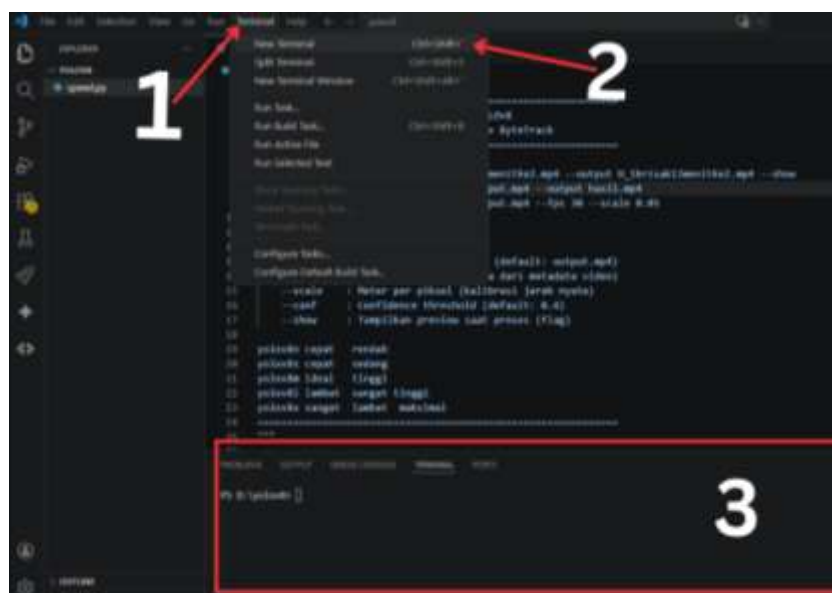


4. Ketik kan program pada file speed.py

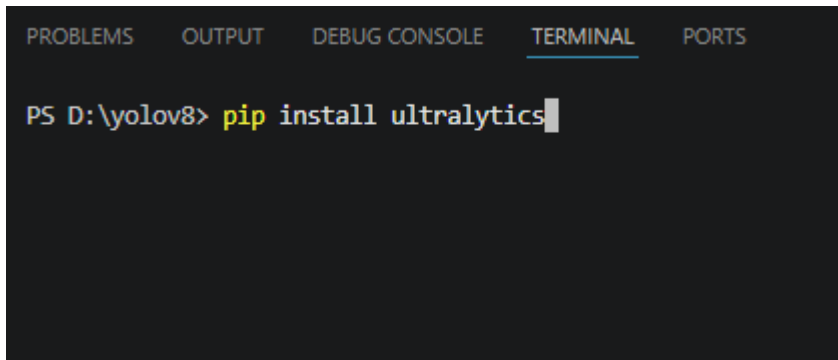


Install library yang dibutuhkan

1. Buka terminal di virtual studio code



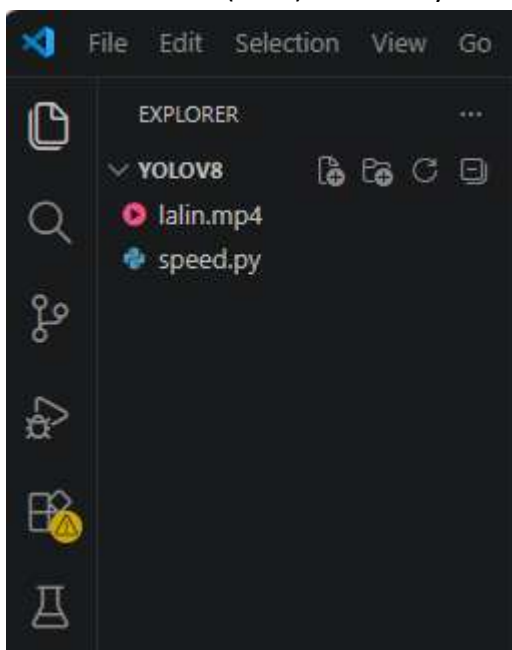
2. Ketik perintah berikut pada terminal
 - a. `pip install opencv-python`
 - b. `pip install numpy`
 - c. `pip install ultralytics`



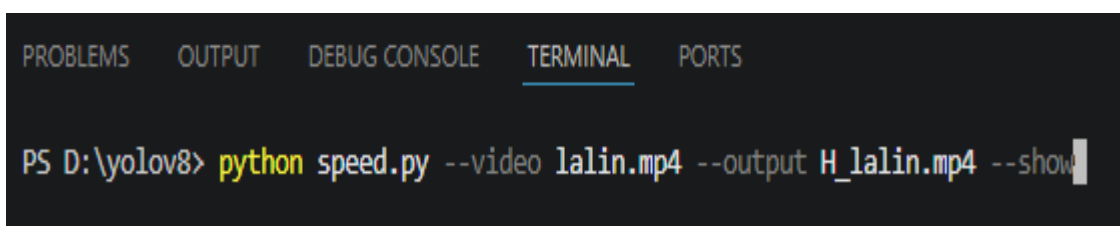
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS D:\yolov8> pip install ultralytics
```

Cara menjalankan program

1. masuk file video (MP4) ke folder yolov8



2. ketik perintah berikut di terminal untuk menjalankan program
 - `python speed.py --video lalin.mp4 --output H_lalin.mp4 --show`



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS D:\yolov8> python speed.py --video lalin.mp4 --output H_lalin.mp4 --show
```

- lalu tekan enter
3. tunggu hingga proses nya 100%

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\yolov8> python speed.py --video lalin.mp4 --output H_lalin.mp4 --show
=====
Deteksi Kecepatan Kendaraan - YOLOv8
=====
Input   : lalin.mp4
Output  : H_lalin.mp4
Skala   : 0.05 m/piksel
Conf    : 0.4

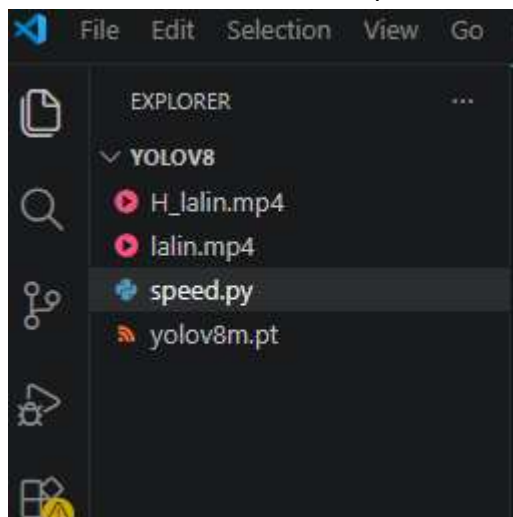
[1/3] Memuat model YOLOv8n ...
      Model siap.

[2/3] Membuka video ...
      Resolusi : 1920x1080
      FPS      : 30.0
      Frames   : 36536

[3/3] Memproses video ...

      Frame    100/36536 ( 0.3%) 30.4s
```

4. kalau video sudah selesai di proses maka hasilnya akan muncul di folder yolov8



- hasil nya file *H_lalin.mp4*